# MTH1001

## UNIVERSITY OF EXETER

## COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

## MATHEMATICS

## May 2019

### Mathematical Structures

### Module Leader: Professor Nigel Byott

### Duration: 2 HOURS.

**Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).**

*Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.*

This is a **CLOSED NOTE** examination.

## SECTION A

1. (a) State the definition of convergence for a sequence $(a_n)$ of complex numbers to a limit $a$, that is, for $a_n \to a$ as $n \to \infty$.

   Show from this definition that if $(a_n)$ and $(b_n)$ are convergent sequences, and $a_n \to a$ and $b_n \to b$ as $n \to \infty$, then $a_n + b_n \to a + b$. (13)

   (b) In each of the following cases, give an example of a series $\sum_{n=1}^{\infty} a_n$ with the property stated. Briefly justify your answers.

   (i) $a_n \to 0$ as $n \to \infty$ but $\sum_{n=1}^{\infty} a_n$ diverges;
   (ii) $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} |a_n|$ diverges;
   (iii) $\sum_{n=1}^{\infty} a_n$ converges but $|a_{n+1}| > |a_n|$ for infinitely many $n$. (12)

   (c) Let $V = \mathbb{R}^4$. Consider the following vectors in $V$:

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \end{pmatrix},$$

$$\mathbf{v}_4 = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 17 \\ 6 \\ 8 \\ 14 \end{pmatrix}.$$

   (i) Find a maximal linearly independent set from the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$. (8)
   (ii) Decide whether $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$ form a basis of $V$. (2)
   (iii) Determine whether the vector $\mathbf{b}$ can be written as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$, and, if it can, write $\mathbf{b}$ in this way. (3)

   (d) Let $V$ be a vector space over $\mathbb{R}$.

   (i) State the definition of a basis in $V$. (2)
   (ii) Let $S$ be a finite linearly independent set of vectors in $V$, and let $T \subset S$ be a subset. Show that $T$ is a linearly independent set of vectors in $V$. (5)
   (iii) Let $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a spanning set of vectors in $V$, where $n = \dim V$. Prove that $\mathbf{v}_1, \ldots, \mathbf{v}_n$ form a basis of $V$. (5)

   [**50**]

## SECTION B

2. (a) (i) State the Completeness Property for the real numbers. (2)

(ii) Let $(a_n)$ be an increasing sequence of real numbers which has an upper bound. By applying the Completeness Property to a suitable set of real numbers, prove that $(a_n)$ converges. (5)

(iii) Hence or otherwise, show that if $(b_n)$ is a decreasing sequence of real numbers which has a lower bound, then $(b_n)$ converges. (5)

(b) Let $\mathbb{R}^\infty$ be the $\mathbb{R}$-vector space of sequences $(a_n)$ with entries in $\mathbb{R}$. Which of the following sets of sequences are subspaces of $\mathbb{R}^\infty$? Justify your answers.

(i) $U_1 = \{(a_n) \in \mathbb{R}^\infty : \exists N \in \mathbb{N} \text{ with } a_n = 0 \ \forall n > N\}$;

(ii) $U_2 = \{(a_n) \in \mathbb{R}^\infty : na_n \to 0 \text{ as } n \to \infty\}$;

(iii) $U_3 = \{(a_n) \in \mathbb{R}^\infty : (a_n) \text{ is not convergent}\}$. (13)

**[25]**

3. (a) What does it mean for a series $\sum_{n=1}^{\infty} a_n$ to converge to a limit $A$? Show that if $\sum_{n=1}^{\infty} a_n$ is convergent then $a_n \to 0$ as $n \to \infty$. (7)

(b) By expressing $2/n(n+2)$ in partial fractions, find a formula for the $n$th partial sum of the series

$$\sum_{n=1}^{\infty} \frac{2}{n(n+2)}.$$

Hence show that this series converges, and find its limit. (6)

(c) Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a spanning set of vectors in a vector space $V$ and let $f$, $g : V \to V$ be linear maps such that $f(\mathbf{v}_i) = g(\mathbf{v}_i)$ for $1 \leq i \leq n$. Prove that $f$ and $g$ are equal. (12)

**[25]**

4. (a) The exponential, cosine and sine functions for complex numbers are defined by the power series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad \cos(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}, \quad \sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}.$$

(These series are absolutely convergent for all $z \in \mathbb{C}$.)

(i) Using these power series, express $e^{iz}$ and $e^{-iz}$ in terms of $\cos(z)$ and $\sin(z)$. (4)

(ii) Deduce that for any $z \in \mathbb{C}$ we have

$$\cosh(iz) = \cos(z) \text{ and } \sinh(iz) = i\sin(z)$$

where the hyperbolic cosine and sine functions are defined for any complex number $w$ by

$$\cosh(w) = \frac{1}{2}(e^w + e^{-w}) \text{ and } \sinh(w) = \frac{1}{2}(e^w - e^{-w}).$$

(8)

(b) Let $V$, $W$ be vector spaces over $\mathbb{R}$ with $\dim V = n$, $\dim W = m$.

(i) Give the definition of a linear map $f : V \to W$. (2)

(ii) Let $f : V \to W$ be an injective linear map. Prove that $m \geq n$. (6)

(iii) Let $f : V \to W$ be a surjective linear map. Prove that $m \leq n$. (5)

[**25**]